Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Construct a relational model for a Library Management System. Identify and define at least three entities involved in this system, and explain how they are related through relationships.	**2**	(CO1, L3)
2)	Describe the key steps involved in the requirement elicitation process and explain their importance for ensuring software quality.	**2**	(CO2, L2)
3)	Analyze the different layers involved in software architecture and explain why each layer is crucial for effective system design.	**2**	(CO3, L4)
4)	Examine the steps involved in requirement elicitation and explain how they ensure the quality of the final product.	**2**	(CO1, L4)
5)	Find and explain two common issues that may arise during the requirement elicitation phase.	**2**	(CO2, L3)

Section B **5*4=20**

6)	Construct and explain DDL and DML queries to create a database for an online bookstore, including tables for books, authors, publishers, customers, and orders. Provide examples of how these tables can be linked through foreign keys.	**5**	(CO3, L3)
7)	Construct a relational database model for a Library Management System. Include at least three entity types with their attributes and explain how relationships between these entities can be represented in the model. Demonstrate the creation of a view that allows librarians to quickly find all books borrowed by a specific member.	**5**	(CO1, L3)
8) a)	Design a relational model for a Library Management System using entity-relation diagrams. Demonstrate how to create various views in this system that provide different perspectives on the data, such as books available by category or borrowed by a specific member. Explain the benefits and limitations of using these views.	**5**	(CO2, L3)

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OR

b)	Determine and explain the significance of each layer in the software engineering process, using an example to illustrate how data flows through these layers.	**5**	(CO2, L3)
9)	Construct a simple database project using DDL and DML queries for managing employee records. Explain your schema design choices and provide example SQL statements for creating tables, inserting data, and querying specific records.	**5**	(CO3, L3)

Section C **10*2=20**

10)	Discuss the process and importance of requirement elicitation for developing an ethical software product. In your answer, explain how potential issues can arise during this phase and suggest strategies to mitigate them. Additionally, describe the role of quality assurance (QA) in ensuring the quality of requirements gathering.	**10**	(CO1, L2)
11) a)	Examine the relational model by creating a database schema for a Library Management System. Your analysis should cover the identification of entities, their attributes, and relationships between them. Additionally, compare and contrast this relational approach with using views in managing data access and retrieval. Provide examples to illustrate your points.	**10**	(CO2, L4)

OR

b)	Examine the key challenges faced during the requirement elicitation phase and analyze how these challenges can affect the quality of software development projects. Additionally, discuss potential strategies to mitigate these issues.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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